

15.075 Statistical Thinking and Data Analysis

Computer Exercises 2

Due November 10, 2011

Instructions: Please solve the following exercises using MATLAB. One simple way to present your solutions is to copy all of your code and results (plots, numerical answers, etc.) into a Word document, which you should submit in class. There is no online submission. **Do NOT share code!**

Exercise 1

In this exercise, we demonstrate the Central Limit Theorem, where each X_i is exponential.

- Generate 1000 random numbers from the exponential distribution with $\lambda = 6$, and plot them in a histogram. This should give you an idea of what the exponential distribution looks like.
- For $n = 2$, repeat the following 1000 times: Generate a random sample of n numbers from the exponential distribution with $\lambda = 6$.
- Compute the sample mean of the n numbers and standardize it using the true mean and standard deviation of the distribution.
- Make a histogram and normal plot of the 1000 sample means.
- Repeat (b)-(d) for $n = 10, 20$, and 100 . Put all four histograms and all four normal plots in the same window. Comment on their shapes.

Solution

Here is the script and the results for my iterations:

```
clear all;
clc;
close all;

num_samples = 1000;
lambda = 6;
mu = 1/lambda;
sigma = mu;

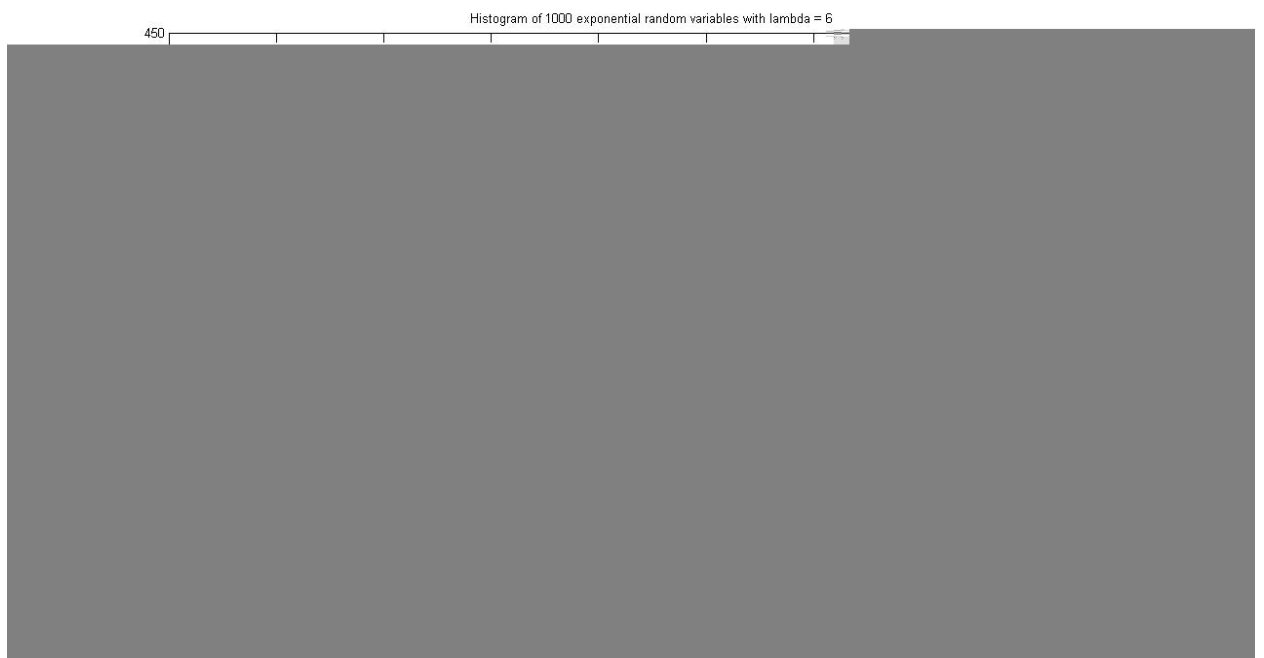
% Part a
% Generate 1000 random numbers from the exponential distribution with
% lambda=6
x = exprnd(mu, num_samples, 1);

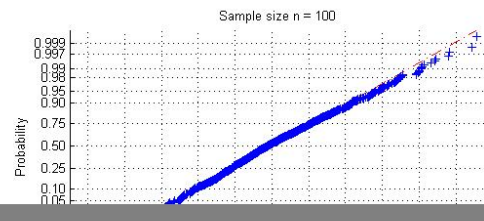
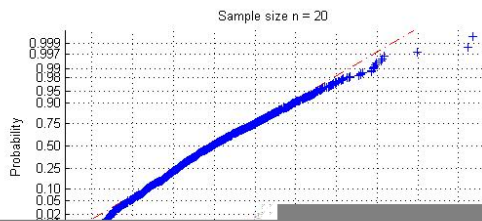
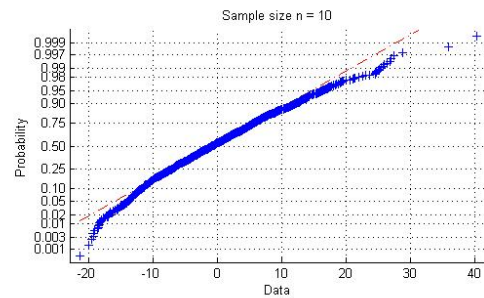
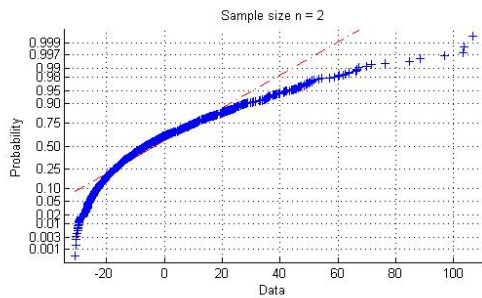
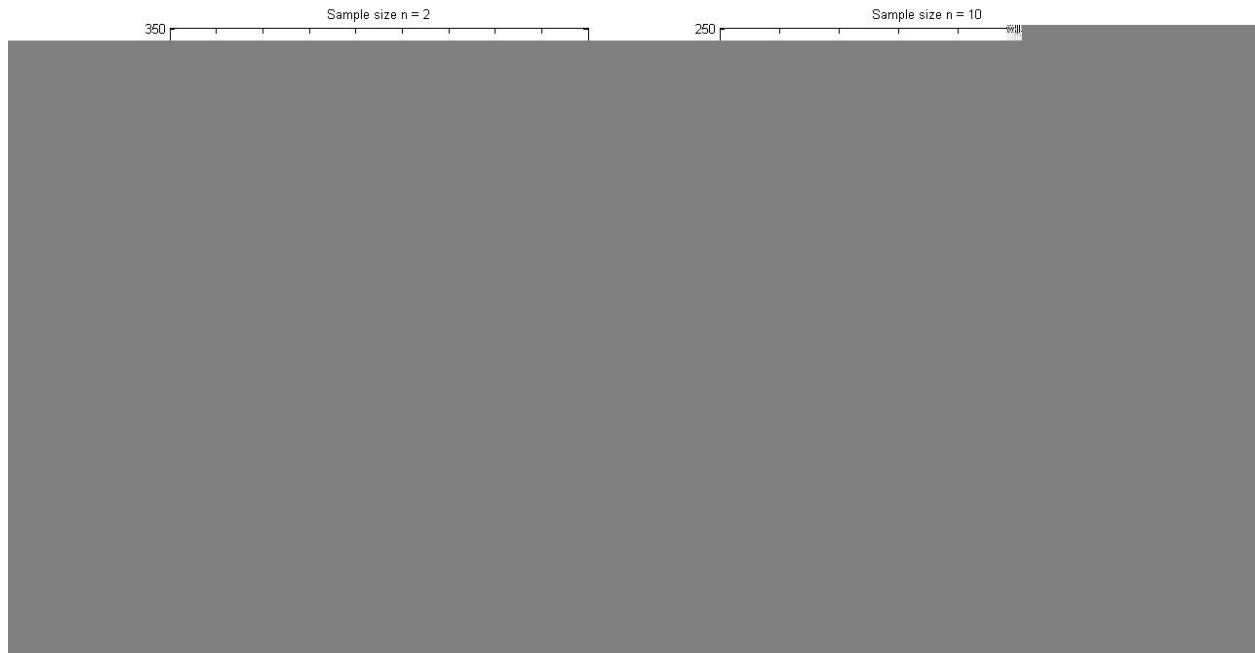
% Plot a histogram
hist(x, 20);
title('Histogram of 1000 exponential random variables with lambda = 6');

% Part b, c, d, e
sample_sizes = [2 10 20 100];
figure;
for i = 1:length(sample_sizes)
    sample_size = sample_sizes(i);
    x = exprnd(mu, num_samples, sample_size);
    sample_mean = mean(x');
    z_score = (sample_mean - mu)/(sigma/sqrt(num_samples));
```

```
% Plot a histogram z-scores
subplot(2,2,i);
hist(z_score);
title(['Sample size n = ', num2str(sample_size)])

% Plot a normal plot of z-scores
subplot(2,2,i);
normplot(z_score);
title(['Sample size n = ', num2str(sample_size)])
end
```





The normal plot for $n=2$ shows that the distribution of sample means is still skewed right since the exponential distribution has a very long positive tail. As n increases, the distribution of sample means becomes increasingly symmetric. With $n=100$, the curve in the normal plot is fairly straight, implying the distribution of sample means is approximately normal.

Exercise 2

Solve Problem 6.23 from the book.

Solution

a) Obviously the α -risk is 0.05 from definition. Let's calculate the β -risk of the rule if $\mu=1$.

The β -risk is just $\beta = 1 - \pi(1) = 1 - \Phi\left[-z_\alpha + \frac{(\mu - \mu_0)\sqrt{n}}{\sigma}\right] = 1 - \Phi(-1.645 + 3) = 1 - \Phi(1.355) = 0.0877$.

b) Here is the script and the results for my iterations:

```
clear all;
clc;
close all;

% Parameters
level = 0.05;
mu_0 = 0;
mu = 1;
sigma = 1;
n = 9;
num_samples = 100;
% Reject if sample mean is greater than critical_value
critical_value = mu_0 + norminv(1 - level) * sigma / sqrt(n);

% Generate 100 samples of size n with mean mu_0
x = sigma * randn(num_samples, n) + mu_0;
x_bar = mean(x');
a_risk = sum(x_bar > critical_value) / num_samples;

% Generate 100 samples of size n with mean mu
x = sigma * randn(num_samples, n) + mu;
x_bar = mean(x');
b_risk = sum(x_bar < critical_value) / num_samples;

>> a_risk

a_risk =

    0.0500

>> b_risk

b_risk =

    0.0700
```

The results are close to their theoretical values.

Exercise 3

Solve Problem 6.25 from the book.

Solution

Here is the script and the results for my iterations:

```
clear all;
clc;
close all;

sigma = 1000;
n = 10;

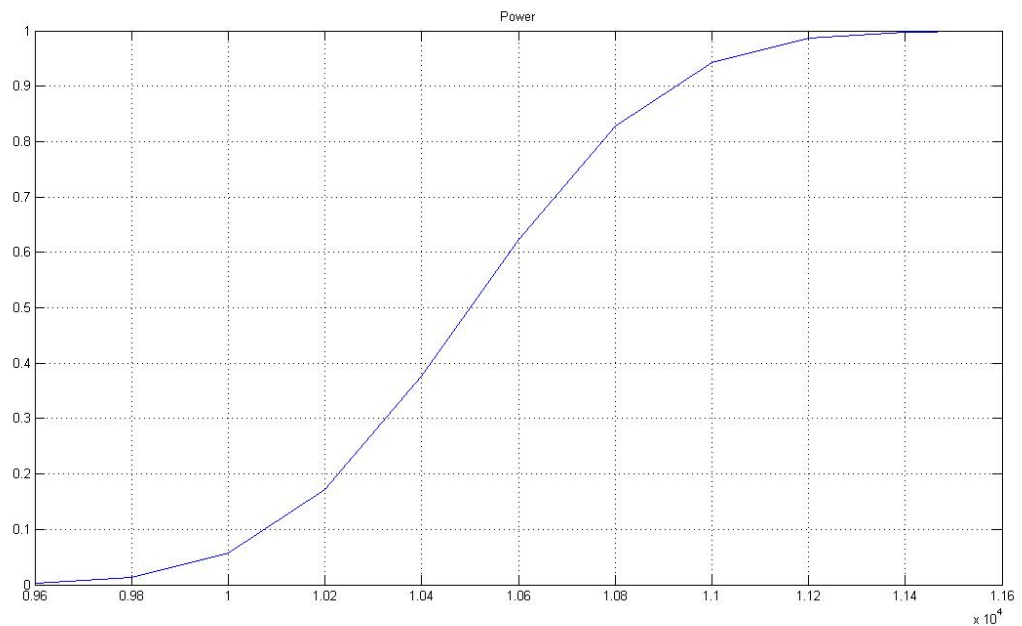
% We reject when x_bar is greater than the critical value
critical_value = 10500;

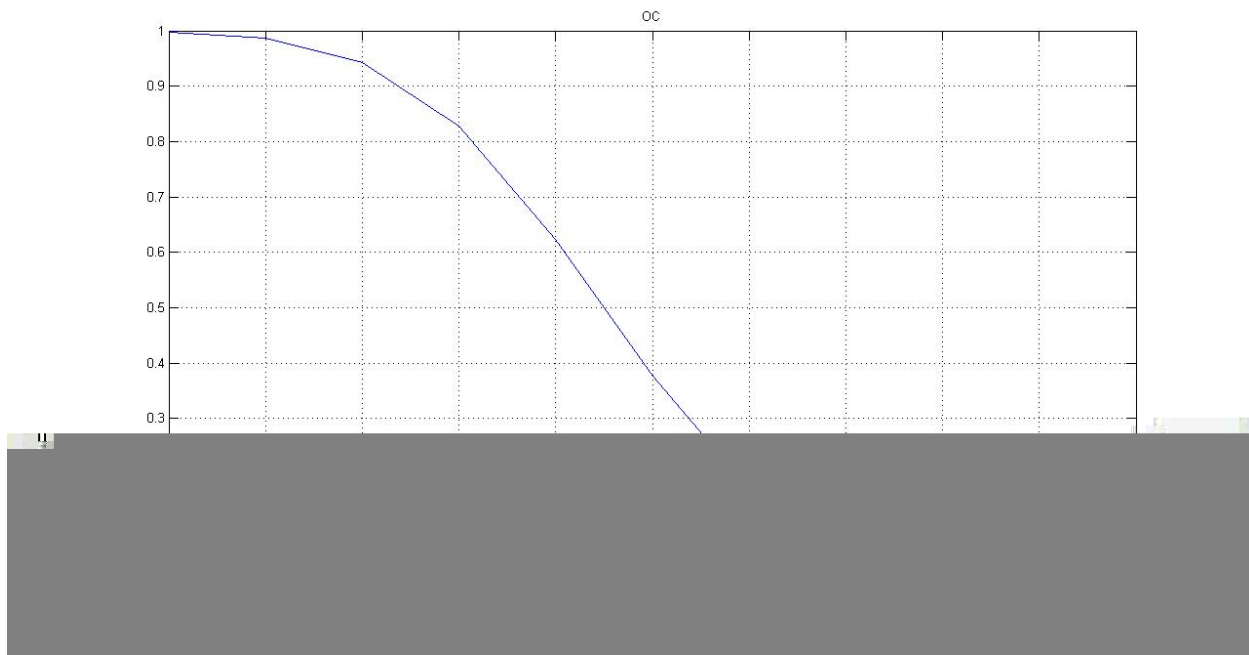
mu = 9600:200:11600;

power = 1-normcdf((critical_value-mu)*sqrt(n)/sigma);
oc = 1-power;

plot(mu,power);
grid;
title('Power');

figure;
plot(mu,oc);
grid;
title('OC');
```





Exercise 4

A thermostat used in an electrical device is to be checked for the accuracy of its design setting of 200°F. Ten thermostats were tested to determine their actual settings, resulting in the following data:

202.2	203.4	200.5	202.5	206.3	198.0	203.7	200.8	201.3	199.0
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Assume the settings come from a normal distribution. Using $\alpha = 0.05$, perform a hypothesis test to determine if the mean setting is greater than 200°F. What are the null and alternative hypotheses? Which test do you use and why? Explain your conclusion using

- An appropriate confidence interval.
- A critical value from the distribution of the test statistic.
- A p -value.

Solution

The hypotheses are

$$H_0: \mu = 200.$$

$$H_1: \mu > 200.$$

We use a t -test since we do not know the true variance, and the sample size is small.

```
clear all;  
clc;  
close all;
```

```
x = [202.2, 203.4, 200.5, 202.5, 206.3, 198.0, 203.7, 200.8, 201.3, 199.0];  
mu = 200;  
alpha = 0.05;
```

```
[h p ci stats] = ttest(x,mu,alpha, 'right')
```

h =

1

p =

0.0227

ci =

200.3729 Inf

stats =

tstat: 2.3223

df: 9

sd: 2.4102

For all the following reasons, we reject H_0 .

- a. The lower 95% CI is $[200.3729, \infty)$, which does not contain 200.
- b. The observed value of the t -test statistic is 2.3223, which is larger than $t_{9,0.05} = 1.833$.
- c. The p -value is 0.0227, which is smaller than the significance level $\alpha = 0.05$.

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15.075J / ESD.07J Statistical Thinking and Data Analysis
Fall 2011

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